

In a particular locality of a state the vehicle users takes a round about route to reach at ~~udden~~ certain places due to presence of a river.

It results in excessive travel time & increased fuel cost. So the state government is planning to construct a bridge across the river. The estimated initial investment for the construction of the bridge will be 40 lakhs. The annual operation and maintenance cost is ₹ 1,50,000. Life of the bridge is 15 years. The value of fuel saving due to the construction of the bridge is 6 lakhs in the first year and thereafter it increases by ₹ 50,000 every year till the end of the life of the bridge. Check whether the project is justified by Benefit Cost Ratio at $i = 12\%$ compound annually.